

MASTER'S THESIS IN PHYSICS

Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the lpGBT ASIC

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Abstract

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1 Introduction

- DT cosmic test stand in Aachen
- Scintillation detector self-developed at institute
- Read out both detectors together
- Create interface to DT chamber to later be used for trigger scintillators (for studies with DT chamber) or other detectors (using DT chamber triggerless reconstructed muons as reference)
- Requirement: Synchronous operation in ns range
- Use lpGBT chip to develop own readout system to interface Phase-2 DT readout
- lpGBT: Commonly used in HL-LHC upgrade project, ensures synchronous operation

2 Theoretical background

2.1 The Standard Model of particle physics

The Standard Model of particle physics (SM) is the state-of-the-art theory model used in particle physics, describing elementary particles and their interactions [1–3]. It successfully predicts the results of many particle physics experiments to a very precise level at the reachable energy scales [4]. The following sections should provide a brief overview of the particles and fundamental forces in the SM. The predicted particles in the SM are shown in fig. 2.1. Despite its success, the current model is considered incomplete [3], which will be discussed briefly as well.

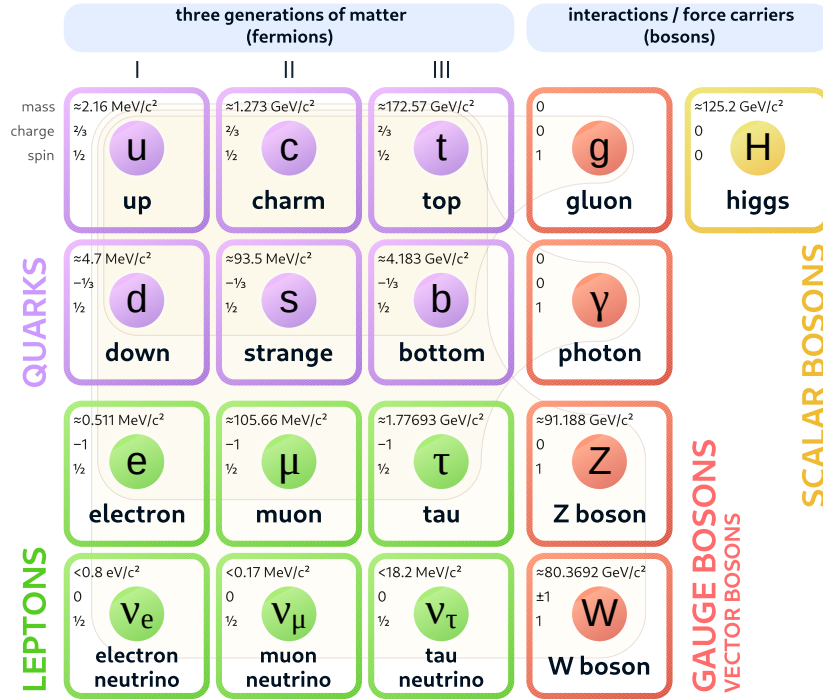


Fig. 2.1: Illustration of the particle content of the SM [5, adapted].

2.1.1 Fermions

All matter particles in the SM are fermions which have half-integer spin and obey Fermi-Dirac statistics [6, 7]. The species of a matter particle is referred to as flavor. The SM predicts three generations of spin- $\frac{1}{2}$ fermions, with four particles in each generation. Per generation there are two quarks (one up-type quark with electric charge $\frac{2}{3}e$ and one down-type quark with electric charge $-\frac{1}{3}e$) and two leptons (one charged lepton with electric charge $-e$ and one electrically neutral and massless neutrino)¹. While the particle masses increase with increasing fermion generation, other properties of the quarks and leptons from different generations are similar.

¹ $e = 1.602 \times 10^{-19} \text{ C}$ is the elementary charge, i.e. the absolute value of the electron charge [8].

The first generation fermions form stable matter, i.e. atoms consisting of protons, neutrons² and electrons, while higher generation fermions decay subsequently into lighter particles of the first generation. For all matter particles, there exists a corresponding antiparticle with similar properties except inverted charge (inverted sign of quantum numbers). Antiparticles are denoted with a bar over the particle symbol or their electric charge (with the index $^\pm$) [1, 3].

2.1.2 Bosons

There are three fundamental interactions in the SM: Electromagnetic interaction, weak interaction and strong interaction. These fundamental interactions are mediated by integer spin gauge bosons which obey Bose-Einstein statistics [9]. In fact, all four mediator particles in the SM have spin-1. The electromagnetic interaction is mediated by the photon (γ) and affects all particles which carry electric charge. The photon itself is massless and electrically neutral, giving the electromagnetic interaction an infinite range. The weak interaction is mediated by the massive $W^\pm$ and Z^0 bosons and has a short range due to the large masses of the mediator particles. It is the only interaction to allow flavor changes of particles. Electromagnetic and weak interaction were unified into the electroweak theory. The strong interaction is mediated by the massless gluon (g). All particles carrying color charge engage in strong interactions, which are the quarks and the gluon itself. While this interaction is the strongest one, it is also restricted in range due to an effect referred to as color confinement. This effect arises from the self-coupling of the gluon and implies that one expects only color neutral bound states and no free quarks in nature [1, 3]. In the electroweak theory, the force mediator bosons are initially massless. However the weak mediators are massive which was both experimentally measured but also required in order for weak interactions to be limited in range. The proposed Higgs mechanism [10, 11] resolves this issue employing spontaneous symmetry breaking. The mechanism leads to massive $W^\pm$ and Z^0 bosons and a massless photon while predicting the existence of an additional spin-0 boson, the Higgs (H) boson [3]. This particle was ultimately discovered in 2012 at CERN [12, 13].

2.1.3 Incompleteness and challenges

While all particles predicted by the SM have now been observed [1] and the model succeeds to predict their interaction properties to very high precision [4], the model is considered deficient. Firstly, the fourth fundamental interaction observed in nature, gravitation, is not included in the model. While gravitativ interaction is significantly weaker than the three interactions of the SM and therefore not relevant for high-energy physics experiments, the model remains incomplete [1]. Secondly, the observation of neutrino oscillations [3, 14] implies non-zero neutrino masses³, however in the SM the neutrinos are considered massless. Cosmic observations hint the existence of dark matter and dark energy, making up $\approx 95\%$ of the matter-energy density of the universe [3, 15]. However the SM only attempts to model the $\approx 5\%$ of baryonic matter. There are additional shortcomings of the SM as the unsatisfactory large number of parameters (26 parameters chosen from experimental observations) and the unresolved matter-antimatter asymmetry observed in nature [3]. This incomplete list should point out the motivation future developments in particle physics theory and experiments.

2.2 Properties of the muon

This thesis focuses on the development of a readout system for a muon detector. Therefore the properties of the particle of interest should be presented in this section in short.

²Protons and neutrons are composite particles of first generation quarks. Note that the free neutron is not a stable particle, however when bound in a nucleus it does not decay.

³Strictly speaking, not all neutrinos are required to be massive, since the oscillations only imply non-zero mass differences between the neutrino flavors.

The muon (μ^-) is the second generation charged fermion. It has an electric charge of $-1e$ and a mass of $m_\mu = 105.658 \text{ MeV}/c^2$. It is therefore $\approx 207\times$ heavier than its first generation relative, the electron (e^-). Since the muon is lighter than the lightest existing hadron (the π^0 meson), it can only decay leptonically [8]:⁴

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu \qquad \mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

The muon lifetime is around $\tau_\mu = 2.196 \mu\text{s}$ [8]. Due to time dilation, high-energy muons can travel significant distances (e.g. in Earth's atmosphere, as discussed in sec. 2.3).

The muon was initially discovered in cloud chamber experiments with cosmic rays in 1937, where a charged particle more massive than the electron other than the proton was postulated [16, 17]. However, it was initially labeled " μ -meson" and confused with the charged pions ($\pi^\pm$ mesons). These mesons have masses in the same order of magnitude as the muon and were thought to be the mediator particle in Yukawa's theory of strong interactions [18]. The leptonic nature of the muon and the distinction from the pions was only revealed later, when a decay of pions into muons could be observed [19].

2.3 Atmospheric muons

Since in this work the muon detector and its developed readout is only operated with atmospheric muons rather than with some other particle source, the creation and properties of atmospheric muons should be briefly described in this section.

2.3.1 Cosmic radiation

Cosmic rays are particles reaching the Earth from space. The first indication of cosmic radiation was through balloon flight experiments of Hess in 1912, who discovered an increasing intensity of ionizing radiation with altitude [20, 21]. Cosmic rays mainly consist of protons, alpha particles and heavier nuclei of a wide range of energies [22–24]. These particles can have many different origins, ranging from our solar system to our galaxy to extragalactic sources. The understanding of cosmic ray sources is still an active field of research today [23, 25]. The flux of cosmic rays is highest for particle energies $\mathcal{O}(1 \text{ GeV})$ but the tail towards high energies extends beyond $\mathcal{O}(1 \text{ ZeV})$ [8]. The energy spectrum of cosmic rays is observed to approximately follow a power law with an energy-dependent spectral index γ [8]:

$$I(E) \propto E^{-\gamma}, \quad \text{where } \gamma \approx \begin{cases} 2.7, & \mathcal{O}(1 \text{ GeV}) \leq E \leq \mathcal{O}(1 \text{ PeV}) \\ 3.0, & \mathcal{O}(1 \text{ PeV}) \leq E \leq \mathcal{O}(100 \text{ PeV}) \\ 3.3, & \mathcal{O}(100 \text{ PeV}) \leq E \leq \mathcal{O}(1 \text{ EeV}) \\ 2.5, & \mathcal{O}(1 \text{ EeV}) \leq E \end{cases}$$

Interactions of the primary cosmic rays with Earth's atmosphere lead to energy loss of the cosmic particles and to a change in particle composition. In strong interactions, secondary particles can be produced. The most common secondary particles are pions ($\pi^\pm$, π^0), however there are also many other particles produced, such as kaons ($K^\pm$, K^0), hyperons and nucleons [24]. Notably, cosmic rays can create large air showers in the atmosphere through cascades of interactions. This was first observed by Auger in 1939 through coincidences of particle detectors on the surface at far distances [26].

⁴Decays with an associated photon or an additional electron-positron pair (e^+e^-) are possible but strongly suppressed with branching fractions $\leq \mathcal{O}(10^{-5})$. Therefore shown decay mode is by far the dominating one [8].

2.3.2 Atmospheric creation of muons

Although their fraction in primary cosmic particles from space is negligible, muons have the largest fraction of charged particles arriving on Earth's surface [22]. These muons originate from the decay of the secondary particles created in Earth's atmosphere. Charged pions and kaons have lifetimes of $\mathcal{O}(10^{-8} \text{ s})$ [8] which is too short to reach Earth's surface, even when accounting for time dilation. The charged pions decay to muons with a branching fraction of ≈ 0.999 [8]:⁵

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu \qquad \pi^+ \rightarrow \mu^+ + \nu_\mu$$

Similarly the charged kaons can decay to muons with a branching fraction of ≈ 0.635 [8]:⁶

$$K^- \rightarrow \mu^- + \bar{\nu}_\mu \qquad K^+ \rightarrow \mu^+ + \nu_\mu$$

Most muons are produced at altitudes of $\approx 10 - 15 \text{ km}$ and lose $\approx 2 \text{ GeV}$ before reaching the surface [22, 27].

2.3.3 Energy spectrum and angular distribution

Needs to be reworked and extended

In the following, the energy spectrum and angular distribution of cosmic muons on Earth's surface should be briefly presented, as they are the baseline to calculate the expected detector response for the measurements presented in this thesis. In principle, calculations of these quantities is quite complex, due to the variety of decay processes and energy loss mechanisms that can occur. However there are useful parametrizations of the measured muon spectra which allow for a simplified model to be used for simulations or approximate predictions [27].

A good approximation of the flux of the vertical (zenith angle $\theta = 0$) atmospheric muons is provided by the following formula [27]:

$$I(E, \theta = 0) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1}$$

In this formula, E_0 is the energy loss of the muon from the point of its creation to the surface, and assumed constant. This is of course an approximation, especially for lower muon energies, as the energy loss is not really independent of the particle energy (see sec. 3.1). Before the meson decays into the muon, it also loses energy on its way towards the surface. When the energy of this particle (and consequently of the resulting muon) is higher, the meson can lose more energy before it decays. This effect is accounted for with the factor $\epsilon > 1$ [27].

The dependence of the muon flux on the azimuth angle ϕ is small and will therefore be neglected here. However there is a dependence on the zenith angle θ which can be parametrized as follows [27]:

$$I(E, \theta) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1} D(\theta)^{-(n-1)}$$

$$\text{where } D(\theta) = \sqrt{\left(\frac{R^2}{d^2} \cos^2 \theta + \frac{2R}{d} + 1\right)} - \frac{R}{d} \cos \theta$$

This formula takes into account the variation in muon path length to the surface for different zenith angles assuming a spherical Earth. $R = 6378 \text{ km}$ is the radius at the Earth's surface [8]⁷

⁵There are also suppressed decay modes with associated photons [8].

⁶There are also semileptonic decay modes with associated pions with a branching fraction of ≈ 0.033 . As before, decays with associated photons are also possible but strongly suppressed [8].

⁷The given value is the nominal Earth equatorial radius and should serve as an approximation for the spherical Earth assumption of the formula.

and $d = \mathcal{O}(10 \text{ km})$ is the vertical path length that the muon travels [22, 27]. When approximating $R \gg d$ (which essentially translates to assuming a flat Earth), one obtains [27]

$$D(\theta) \approx 1/\cos \theta$$

This yields the more commonly known approximation of the atmospheric muon angular dependence with a cosine power law with typical values of $n \approx 3$ (also see sec. 2.3.1) [27]:

$$I(E, \theta) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1} \cos^{(n-1)} \theta \quad (2.1)$$

Using available values for the parameters of the presented approximation, the expected muon rate which is measured in this thesis should now be estimated. The available parameters in [27] are obtained from data measured in Hamburg, Germany (53.56° N, 10° E, altitude: m) which is considered reasonably close to Aachen (53.56° N, 10° E, altitude: m) to use similar values. The obtained energy and zenith angle dependency are plotted in fig. 2.2.

$\int_{1 \text{ GeV}}^{\infty} dE I(E, \theta = 0)$	n	E_0	ϵ
$(65.2 \pm 1.5) \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$	3.00 ± 0.02	$(23.78 \pm 0.30) \text{ GeV}$	2000 (fixed)

Tab. 2.1: Obtained values for a fit the approximation in eq. 2.1 to atmospheric muon data taken at Hamburg, Germany. Taken from [27].

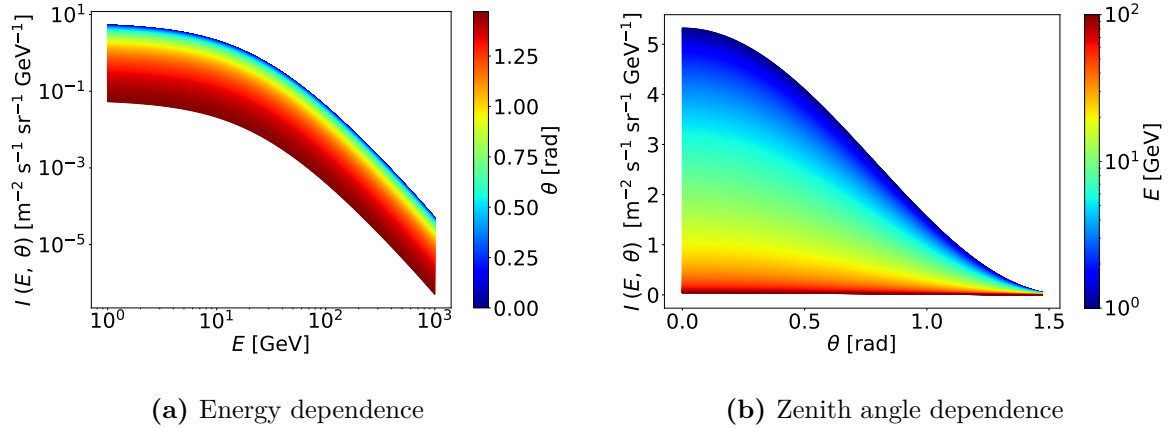


Fig. 2.2: Distributions for the atmospheric muon energy (E) and zenith angle (θ) dependence according to eq. 2.1 for the parameters in tab. 2.1.

The expected muon rate for a detector with area A can now be obtained by integration over energy, solid angle and area

$$\dot{N} = \int_{E_{\text{thres}}}^{\infty} dE \int d\Omega \int dA I(E, \theta)$$

For this calculation, the energy detection threshold E_{thres} should be taken into account. Note that the solution of the solid angle and area integrals is not trivial for arbitrary detector geometries, since the effective detector area might be a function of the solid angle. Typically one divides

3 Principles of particle detection

3.1 Interaction of particles and matter

Particle detector detects the effects due to the particle interaction, not the particle itself.

3.1.1 Charged particle interactions

3.1.1.1 Non-ionizing collisions

3.1.1.2 Loss through ionization

3.1.1.3 Emission of Bremsstrahlung

3.2 Scintillators and photon detection

- Scintillator: Photon emission due to energy deposition of particle
- Sensitive photon detection: PMT, SiPM (discussed in detail)

3.2.1 Scintillators

- Organic and anorganic scintillators
- Plastic scintillator (that is used in this thesis)

3.2.2 Silicon photomultipliers

- Geiger-Avalanche Photodiode array
- Sensitivity and performance
- Output signal shape
- Characteristic noise spectrum

3.3 Gaseous detectors

3.3.1 Operation modes of gaseous detectors

- Ionization chamber
- Proportional mode
- Nonlinear amplification mode
- Geiger mode

3.3.2 Drift chamber

- Proportional mode
- Constant electric field and drift velocity
- Correlation of charge travelling time and initial ionization position
- Drift chamber legacy in Aachen

3.4 Modern readout electronics

3.4.1 General aspects of detector readout

- Quantity to read out: E.g. voltage / current
- Discrimination of hits by amplitude
- Good time resolution
- More advanced: Resolution of signal amplitude and length, large number of channels

3.4.2 FPGA devices

- Brief history of FPGAs
- Fully programmable logic (difference to CPU)
- In particle physics: Fast, synchronized readout and processing of many channels

3.4.3 Basics of logic circuit development

- Description of logic circuitry implemented on FPGA
- Combinatorial logic
- Sequential / clocked logic
- Hardware Description Language
- IP cores

4 CMS Drift Tube system

4.1 Compact Muon Solenoid experiment

4.1.1 Large Hadron Collider

- Image of CERN accelerator complex
- Key facts: Luminisity, collision center-of-mass energy

4.1.2 Experiment overview

- CMS rendering
- Brief listing of all subdetectors

4.2 Drift Tube system

4.2.1 Subdetector setup

- DT chambers in CMS: Wheels, sectors, stations
- Components of one DT chamber: 2-3 superlayers, 4 layers, drift cells
- Gas mixture, HV values

4.2.2 Detector impact and performance

- Relevance of muon detection (e.g. dimuon Higgs decay)
- DT performance (efficiency and active channels)
- DT system resolution

4.3 Phase-2 DT readout electronics

4.3.1 Motivation for upgrade

- High-Luminosity LHC
- Increased pileup and tightened radiation environment

4.3.2 Minicrate electronics

- Minicrate location on chamber
- Minicrate functionalities
- On-Board Electronics for Drift Tubes boards

4.3.3 HTG-940 backend

- Prototype backend (AMD Xilinx Virtex UltraScale+ FPGA)
- Production backend: Serenity and BMTL1 (not available yet)
- Backend firmware (mention own modifications)

4.4 DT test stand in Aachen

4.4.1 MB1 DT chamber

- MB1 DT chamber from production in 2000s
- Reactivated infrastructure

4.4.2 Installation of Phase-2 minicrate

- Minicrate on DT chamber
- Detailed setup and channel mapping

5 Triggerless Drift Tube data analysis and track reconstruction

5.1 Intra-channel timing calibration with testpulses

- Importance of timing calibration in DT
- Refer to CMS DT calibration procedure
- Show testpulse response
- Discuss different testpulse cable length in theta SL
- Show derived timing corrections for individual channels

5.2 Drift cell hits

- Show hit occupancies of all wires
- Give list of masked (noisy) and dead wires
- Discuss rate
- Highlight and discuss feature of secondary hits in range 0-400 ns
- Apply dead time for individual channels

5.3 Superlayer hit clustering

- Triggerless reconstruction: Need to consider all recorded hits as potential muon signature
- Cluster hits within maximum drift time window
- Pattern rate discussion
- Compare to rates of fake patterns (to estimate noise influence)

5.4 Superlayer cluster fitting

- Assume constant drift velocity, straight muon track
- Perform linear fit for all laterality combinations with parameters: Arrival time, position in cell, track inclination
- Select laterality combination with best goodness of fit
- Discuss fit results
- Apply appropriate goodness of fit cut

5.5 Superlayer relative time alignment

- Mention relative alignment problem of OBDTs
- Show double peak in time difference between SL fits
- Establish relative time correction

5.6 Grouping of superlayer cluster fits

- Clustering algorithm creates multiple clusters in case of additional nearby hits
- Cope this by grouping together clusters within certain time window within same superlayer

5.7 Global track building

- Select cluster groups within time window from all superlayers
- Only use cluster groups with no ambiguity (only one cluster per SL)
- Ensure matching tracks in both phi superlayers
- Calculate 3D track from 2D projections from theta and phi superlayers

5.8 Reconstructed muon tracks

- Discuss muon rate
- Show angular distributions (with solid angle correction)

5.9 Acceptance correction and angular distributions

- Establish geometric simulation of DT chamber with simulated muon tracks
- Derive (rudimentary) geometric acceptance correction from simulation
- Apply acceptance correction to data
- Compare to expected cosmic muon angular distributions

6 SiPM-based scintillation detector

6.1 Detector setup

Mention self-development of detector at institute [28] [29].

6.1.1 Mechanical structure

- 2x 16 plastic scintillator strips with wavelength-shifting fibers
- Dual SiPM readout for each strip

6.1.2 SiPM electronics

- SiPM front-end board with preamplifier
- Backbone of detector with bias voltage supply
- Raspberry Pi for backbone control

6.2 FPGA readout system

- Mention completely new hardware and firmware development in scope of this thesis, based on previous experience and efforts of our group
- AMD Xilinx UltraScale+ ZynqMPSoC
- Control hardware and firmware from Petalinux running on ARM core of ZynqMPSoC
- Show scheme / overview of readout system

6.2.1 Input circuitry

- Synchronous testpulse source
- SiPM signal input and testpulse input (AC coupled)
- Individually selectable threshold for all channels
- (Show screenshot from SPICE simulation)

6.2.2 FPGA firmware overview

Mention fully self-developed readout firmware.

6.2.3 Signal sampling with FPGA-internal comparators

- Comparators for input signal discrimination, however no conventional, dedicated hardware but FPGA-internal comparators used.
- Use FPGA-internal blocks for deserialization to increase time resolution

6.2.3.1 Working principle

- FPGA-internal comparators (intended for differential signalling)
- Novel concept: Use FPGA-internal comparators with analog signals [30] [31] [32]
- Voltage limits of inputs
- Deserializers for higher time resolution

6.2.3.2 Voltage bias calibration

- FPGA-internal comparators (on FPGA silicon die) are subject to production uncertainties
- Unfolds in voltage bias on comparator switching point
- Scan input voltage range while keeping other comparator input constant to determine switching point of comparator (Metastability calibration)
- Account for measured biases when setting thresholds

6.2.3.3 Input timing calibration

- Bank-specific calibration of deserializers on each power-cycle leads to non-deterministic signal delays
- Use simultaneous testpulses to measure these delays
- Use delay module in firmware to correct for these effects already on the device
- Mention ringing of testpulse signal in circuit, but explain that no negative effect on calibration

6.2.4 Online signal processing

6.2.4.1 Timestamp assignment

- Hit arrival time translated to (absolute) timestamp
- Same format as ODBT hits

6.2.4.2 Data funneling

- Limited bandwidth: Can not transmit hits of all channels together
- However do not want to lose hits
- Solution: Introduce individual buffer (FiFo) for all channels
- Dynamically cycle through and activate outputs of different channel buffers
- Reduces possible hit rate compared to no funnel

6.3 lpGBT Mezzanine PCB

- Hardware interface between scintillator readout system and DT backend
- ODBTs use lpGBT as well therefore compatible format
- Synchronous clock (with backend) supplied by lpGBT chip

6.3.1 lpGBT ASIC

- Low-power GigaBit Transceiver ASIC developed centrally by CERN
- Key facts: Data transfer rate, clock recovery
- Optical fiber interface to backend
- Electrical link (eLink) interface to frontend FPGA
- Control interface via optical fiber or wired I2C

6.3.2 Mezzanine board

- Connects eLinks between lpGBT and FPGA
- (Show bit error rates of PRBS7 test)
- Show recovered 40 MHz clock
- Hosts optical transceiver to convert signals to/from optical fiber

6.3.3 Backend firmware modifications

- Modified data flow in DT backend to accomodate reception of data via slow control link from the lpGBT Mezzanine (OBDT has dedicated trigger links for data)
- (Show received counter data by backend)

7 Scintillator data analysis

7.1 SiPM noise scan

- Scan of SiPM rates for different thresholds with developed readout system
- Show resulting curve
- Discuss visible exponential (expected) behavior
- Aliasing effects (too coarse scanning steps) hide expected steps in spectrum

7.2 SiPM hits

- Show single SiPM occupancies
- Discuss observed secondary pulses: Ringing in circuit or SiPM after-pulses

7.3 SiPM coincidences

- Coincidence of two SiPMs of same scintillator strip
- Discuss strip rate
- Show time difference between both SiPMs, discuss width / time localization
- Compare to expected fake coincidence rate

7.4 Scintillator strip coincidences

- Coincidence of two strips of two detector layers (create "pixels")
- Discuss pixel rate
- Show time difference between both strips
- Compare to expected fake coincidence rate

8 Combined measurement with both detectors

8.1 Combined detector setup

- Overview schematic of combined detector and readout setup
- Highlight shared backend and synchronized clocks

8.2 Temporal hit correlation

- Group scintillator pixel hits with reconstructed DT tracks in time window
- Discuss rate of correlated events (compared to scintillator pixel hits before correlation)
- Show time difference between DT and scintillator
- Discuss distribution, highlight the synchronization

8.3 Spatial muon correlation

- Show cumulated reconstructed DT tracks for "matched" hits between scintillator and DT
- Highlight good geometric agreement
- Mention need for alignment analysis (which in principle would be possible with this data)

9 Summary and Outlook

Summary:

- Completely new SiPM readout system developed
- Incorporating modern FPGA data acquisition and FPGA-internal comparators for analog signal discrimination
- Adapted and developed new calibration procedures necessary for FPGA readout system
- Developed lpGBT Mezzanine to interface DT backend
- Developed FPGA firmware for readout and interfacing the lpGBT
- Successful modification of DT backend firmware to interface developed readout system
- Installed and commissioned Phase-2 minicrate with DT chamber at test stand in Aachen
- Framework to analyze DT and scintillator data together
- Implemented own triggerless reconstruction code which follows the CMS DT reconstruction
- Successful correlation of measured DT tracks and scintillator hits in space and time

Outlook:

- Understand open problems
- Implement coincidence IP core (currently still in development)
- Use system to perform various studies with DT chamber, scintillator or even other detectors connected to the developed readout system

Appendix

A Additional details

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List of Acronyms

SM Standard Model of particle physics

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